A Scope Observation Concerning K3 Problem 4.8

The displayed arbitrary-X formulation is already negative after one stabilization

Purpose of this note. This note reports a possible scope issue in Problem 4.8 of *K3: A New Problem List in Low-Dimensional Topology*. The underlying stabilization theorem is not new. The observation is that its direct application appears to give a negative answer to the problem as displayed for arbitrary qualifying X. This does not answer the separate Fintushel–Stern question with X fixed to the K3 surface $E(2)$.

1 The displayed problem and the K3 special case

Problem 4.8 fixes a closed simply connected smooth 4–manifold X and a smoothly embedded torus $T \subset X$ satisfying

$$\pi_1(X - T) = 1, \qquad [T]^2 = 0.$$

For a knot $K \subset S^3$, let X_K denote Fintushel–Stern knot surgery along T. The displayed question asks whether, for prime knots K_1, K_2, a diffeomorphism $X_{K_1} \cong X_{K_2}$ forces K_1 to be K_2 or its mirror.

As written, X is arbitrary subject to the stated hypotheses. In particular, the statement imposes no minimality, irreducibility, or unstabilized condition. Remark (2) then identifies $X = K3 = E(2)$ as a particular case, attributes that case to Fintushel and Stern, and states that beyond equality of Alexander polynomials no further constraint is known. The distinction between the displayed arbitrary-X question and this fixed K3 case is essential.

2 The previously published stabilization theorem

Let Y contain a c-embedded torus, so that the knot-surgery construction is performed in a cusp neighborhood. Akbulut's Theorem 2.2 states that for every knot K,

$$Y_K \# (S^2 \times S^2) \cong Y \# (S^2 \times S^2). \tag{1}$$

The paper notes that Auckly independently observed the same stabilization phenomenon. Baykur later proved a one-stabilization theorem under the more general hypotheses of a square-zero torus with simply connected complement.

Equation (1) is an established theorem. No new stabilization result is claimed here.

3 The direct consequence for the displayed formulation

Proposition 1. *The displayed arbitrary-X formulation of K3 Problem 4.8 has a negative answer.*

Proof. Let $Y = E(2)$ and let $F \subset Y$ be a regular elliptic fiber. The fiber is a c-embedded square-zero torus, and its complement is simply connected. Put

$$Z = S^2 \times S^2, \qquad X = Y \# Z,$$

where the connected sum is performed in a ball disjoint from F. Then X is closed, simply connected, and smooth; the same copy of F has $F^2 = 0$ and $\pi_1(X - F) = 1$.

Knot surgery is disjoint from the connected-sum neck. Regrouping the same cut-and-paste pieces gives an orientation-preserving locality diffeomorphism

$$X_K \cong Y_K \# Z \tag{2}$$

for every knot K. Applying Akbulut's theorem to (Y, F) gives

$$Y_K \# Z \cong Y \# Z = X. \tag{3}$$

Combining (2) and (3),

$$X_K \cong X$$

for every knot K.

Now choose

$$K_1 = T(2,3), \qquad K_2 = T(2,5).$$

Both are prime torus knots. Their Alexander polynomials have respective breadths 2 and 4, so they are neither equivalent nor mirrors. Yet

$$X_{K_1} \cong X \cong X_{K_2}.$$

This is a counterexample satisfying every hypothesis in the displayed arbitrary-X formulation. □

4 What the observation does and does not establish

- It gives a direct negative answer to Problem 4.8 exactly as displayed when arbitrary qualifying ambient manifolds are permitted.
- It does not solve the unstabilized fixed-$E(2)$ question in Remark (2).
- It does not introduce a new 4–manifold theorem: the decisive input is Akbulut's published Theorem 2.2, or alternatively Baykur's later one-stabilization theorem.
- The potentially new point is the explicit application of that old theorem to the scope of the newly published Problem 4.8. A targeted search found no public correction or prior explicit notice of this implication, but no claim of worldwide priority is made without editorial confirmation.

5 Suggested editorial clarification

If the intended problem is the Fintushel–Stern question for the K3 surface, the most direct repair would be to state the ambient pair explicitly, for example:

> Let $X = E(2)$ and let T be a regular elliptic fiber. If K_1 and K_2 are prime knots such that X_{K_1} and X_{K_2} are diffeomorphic, must K_1 be K_2 or its mirror?

More general unstabilized variants may also be meaningful, but additional hypotheses would need to be selected by the authors; minimality or irreducibility should not be inserted without a separate analysis.

The editors and scribes are respectfully asked to clarify:

1. whether the arbitrary-X formulation was intended;
2. whether the implication of Akbulut's Theorem 2.2 was already known in connection with Problem 4.8; and
3. whether a correction restricting the problem to $X = E(2)$, or another explicitly unstabilized class, is appropriate.

6 Verification materials

An explicit certificate, source ledger, independent replay program, Lean 4 formalization of the logical composition, and hash-bound kernel evidence are available at

`https://open.argusbot.cn/results/2884`.

The Lean development checks the composition from the ambient, locality, and stabilization hypotheses to the counterexample witness. It does not formalize smooth 4–manifold topology or Akbulut's theorem from first principles; those inputs remain explicit hypotheses tied to the primary sources.

References

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[2] R. I. Baykur, *Dissolving knot surgered 4-manifolds by classical cobordism arguments*, J. Knot Theory Ramifications **27** (2018), no. 5, 1871001. https://arxiv.org/abs/1704.04491.

[3] R. I. Baykur, R. C. Kirby, and D. Ruberman, *K3: A New Problem List in Low-Dimensional Topology*, AMS Surveys and Monographs, vol. 295, 2026, Problem 4.8. https://doi.org/10.1090/surv/295.

[4] R. Fintushel and R. J. Stern, *Knots, links, and 4-manifolds*, Invent. Math. **134** (1998), 363–400. https://arxiv.org/abs/dg-ga/9612014.